Data Analytics 3

Predictive Analytics and Deep Learning

Prof. Dr.-Ing. Binh Vu

Welcome to Predictive Analytics and Deep Learning!



Course Organization

00

Organization

Lectures

- 16 lectures + 1 day for NVIDIA certificate
- 09:30 - 12:45 and 14:00 – 17:15
- Assignments: In-class
  - Solution: GitHub
- Material: Will be uploaded on Moodle
  - „Analytics 3: Predictive Analytics and Deep Learning"
  - If anyone is not able to access, kindly contact me



Image: https://www.workforcehub.com/

Why Me?

- Prof. Binh Vu explained everything very perfectly. He made a very difficult topic (like Deep Learning) very interesting to study.
- Great learning experience overall
- Hands-on experience was well organized by professor.
- The professors were very approachable and cleared all of my doubts clearly to my understanding.
- the lectures are super fun to learn and interesting too
- i learned very new direction in understanding things and learnt how market is implementing it. it was good
- Professor helped us a lot. He stayed back for us, clear our doubts in theory and practical assignments too. He was very helpful. he taught us well. Every work weekend, we had full day lecture. That day will be so tiring. Deep learning was new to us and learning continuously that one day is too much for brain.
- I really enjoyed this module. The content was well structured, the practical exercises were helpful, and the assignments reinforced the concepts taught in class. The instructor explained the topics clearly, and the module provided valuable real-world knowledge.
- It's properly designed.
- Nothing ,prof bin is great
- Everything was good
- Shit, Binh Vu only cares about attendance and doesn't matter if you do the work and doesn't matter how good your work is.

Why Me?

top 5?

For **overall research prestige and impact**:

1. **NeurIPS** — broad machine learning and AI
2. **ICML** — core machine-learning research
3. **ICLR** — deep learning, foundation models and representation learning
4. **CVPR** — computer vision and multimodal AI
5. **ACL** — NLP and large language models

A recent cross-conference study likewise treats these five as the premier venues spanning modern AI research. arXiv

how many universities of applied sciences' paper accepted?

Worked for 4m 8s ›

For the **2024 editions of NeurIPS, ICML and ICLR**, I found:

2 German universities of applied sciences

with **3 distinct accepted papers**:

Institution	Conference	Paper
Technische Hochschule Ingolstadt (THI)	ICLR 2024	*Würstchen: An Efficient Architecture for Large-Scale Text-to-Image Diffusion Models*
Technische Hochschule Ingolstadt (THI)	ICML 2024	*Detecting Influence Structures in Multi-Agent Reinforcement Learning*
Hochschule Düsseldorf (HSD)	NeurIPS 2024	*MassSpecGym: A Benchmark for the Discovery and Identification of Molecules*

So, among Germany's **209 universities of applied sciences**, only about **1%** had a paper represented at one of these three conferences in that year. This is an affiliation-based count, not an official statistic published by the conferences; the underlying dataset covers author affiliations through 2024 only. destatis.de +1

You have an accepted submission at the ICML conference in 2026. Your submission at OpenReview uses the account binh.vu@srh.de. A matching account did not exist at icml.cc for this email address, so we created one for you. Please read this important email entirely to ensure that your paper/spotlight/etc gets linked to your icml.cc account properly.

Dear Binh Vu,

Congratulations! We are delighted to inform you that your submission to ACL 2026, Benchmarking Vietnamese Legal Knowledge of Large Language Models, has been accepted as a **Findings Paper**.

Dear Binh Vu,

We are delighted to invite you to serve as a Reviewer for the Conference on Neural Information Processing Systems (NeurIPS) 2026. Reviewers are critical to NeurIPS and to the many communities it serves. We sincerely hope you are able to accept our invitation and help make NeurIPS 2026 a success.

Dear Binh Vu,

We are delighted to inform you that your submission has been accepted at NeurIPS 2026 ED Track as a poster. Congratulations!

There were 3,757 valid paper submissions with a PDF to the NeurIPS ED Track this year, of which the program committee accepted 971 (25.8%) papers in total, of which 923 are accepted as posters, 33 as spotlights, and 15 as orals.

Organization

Examination

- Deadline: Monday, 9th November 2026 at 23:59
- Project Work
  - Individual project
  - Submission on Moodle
  - No presentation
- Final Grade = 90 % Exam Grade + 10 % Attendance
  - Attend < 8 lectures: 0% bonus
  - Attend >= 8 lectures: Up to 10% bonus



Image: https://www.indiatoday.in

Requirements

- Collect data and train a deep learning model for **Land Use and Land Cover (LULC) semantic segmentation** that uses **Sentinel-1** data as its only input.
- Build the prototype
  - **Input**: latitude/longitude of the location and a start month.
  - **Processing**: Run the trained model to produce 12 monthly LULC maps.
  - **Calculation**: for each month, give the percentage of the area covered by each class. Also give the month-to-month change and the total change from the first to the last month, both in percentage points.
  - **Visualization**: LULC map of the selected location over 12 months
- Submit
  - The **dataset**: a download link if too big
  - The **notebook(s)**: ensuring it displays the execution results and visualizations of training performance
  - The **prototype**: source code, a README (setup, dependencies, required accounts or API keys) and a requirements file.
  - The trained **model** weight.
  - A short **screen recording** of a full run for at least one urban area of your choice.

Requirements

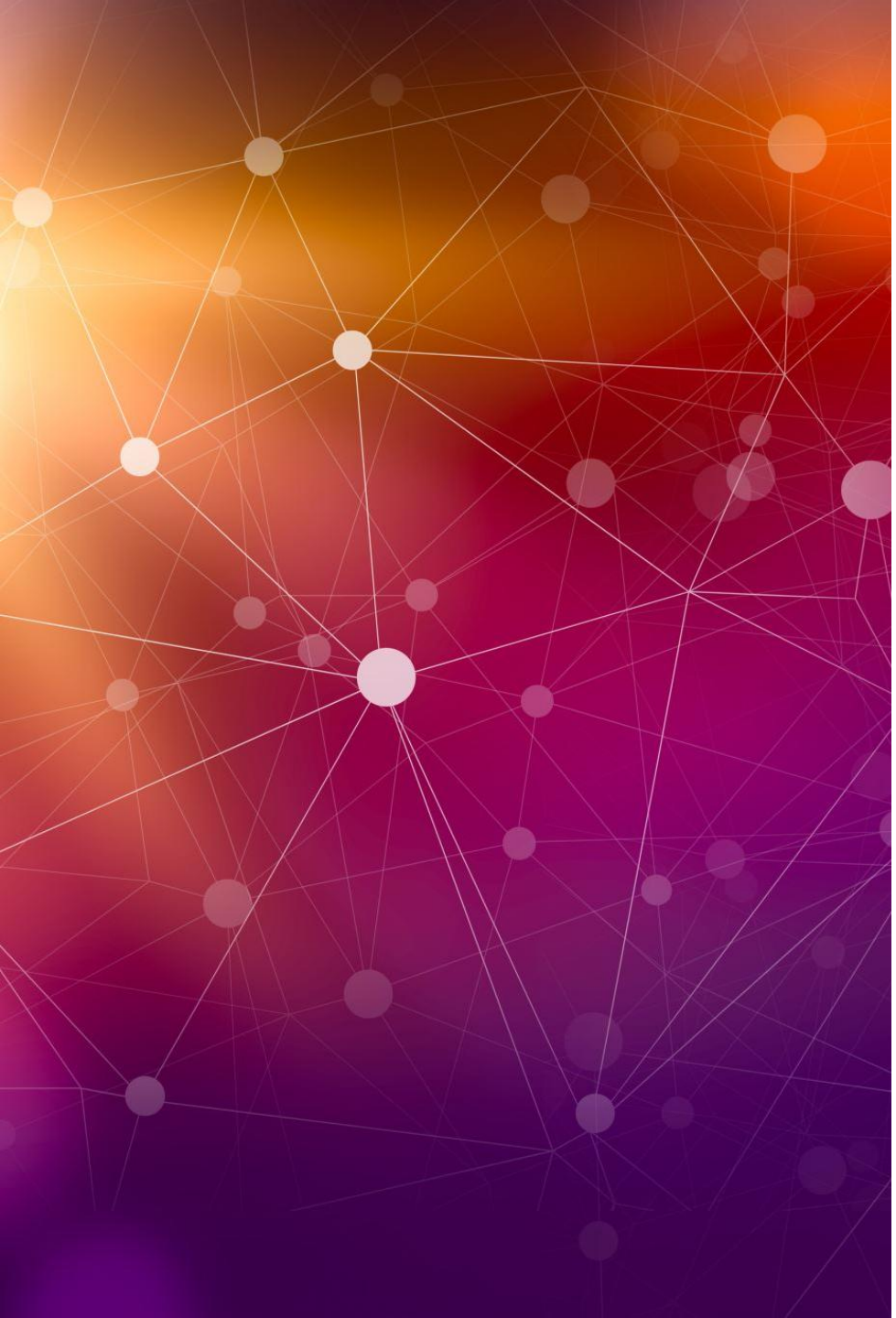
The report must cover:

- **Dataset and preprocessing**: Describe the data source, regions, time period, classes and class distribution. Describe all preprocessing steps, such as:
  - calibration, dB conversion, speckle filtering and terrain correction
  - normalization and patching
  - how labels were matched to the imagery
  - augmentation and data splits
- **Model**: Give a clear explanation of the architecture and training setup, and the reasoning behind your design choices.
  - Experimental results.
  - overall accuracy (OA), mean IoU (mIoU) and per-class IoU/F1 on the test set
  - training curves
  - a comparison of model variants, if you tested more than one
- **Prototype**: Cover
  - the system architecture
  - data retrieval and monthly compositing
  - how class percentages and changes are calculated
  - a case study of your chosen location, with the results sanity-checked against an independent LULC map not used for training
- **Critical discussion**: Cover limitations of the model and the prototype, including
  - performance on regions unlike the training data
  - SAR-specific confusion between classes
  - seasonal effects that can look like real land-use change
  - how classification errors inflate the calculated change percentages

Contact

For any queries or feedbacks or issues regarding the module (or the program):

- **Prof. Dr.-Ing. Binh Vu**
- Room: Arc 215
- Phone: +49 6221 6799 - 208
- Email: binh.vu@srh.de



Course Structure

- Foundation of Deep Learning
- Linear Neural Networks
- Multilayer Neural Networks
- Backpropagation
- Optimizer & Overfitting
- Convolutional Neural Networks
- Transfer Learning
- Recurrent Neural Networks
- Graph Neural Networks



Questions?

Introduction to Deep Learning

01

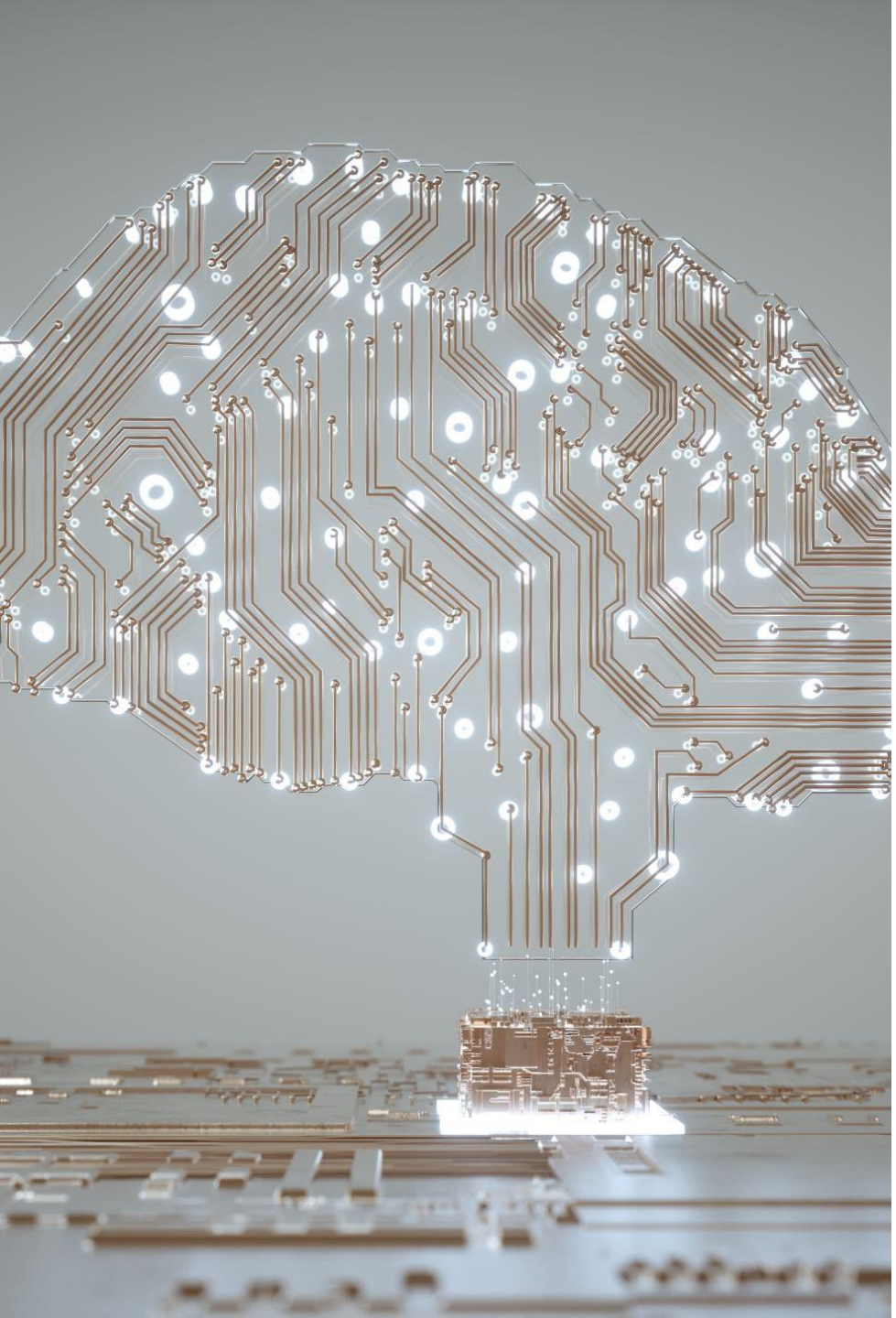


Table of Content

- Artificial Intelligence
- Neural Networks
- The History of Neural Networks
- Deep Learning Characteristics
- Applications
- Deep Learning Frameworks

Artificial Intelligence



Artificial Intelligence?





Image Source:
https://www.instagram.com/short.squeez
https://www.businessinsider.com/amazons-just-walk-out-actually-1-000-people-in-india-2024-4

Artificial Intelligence?

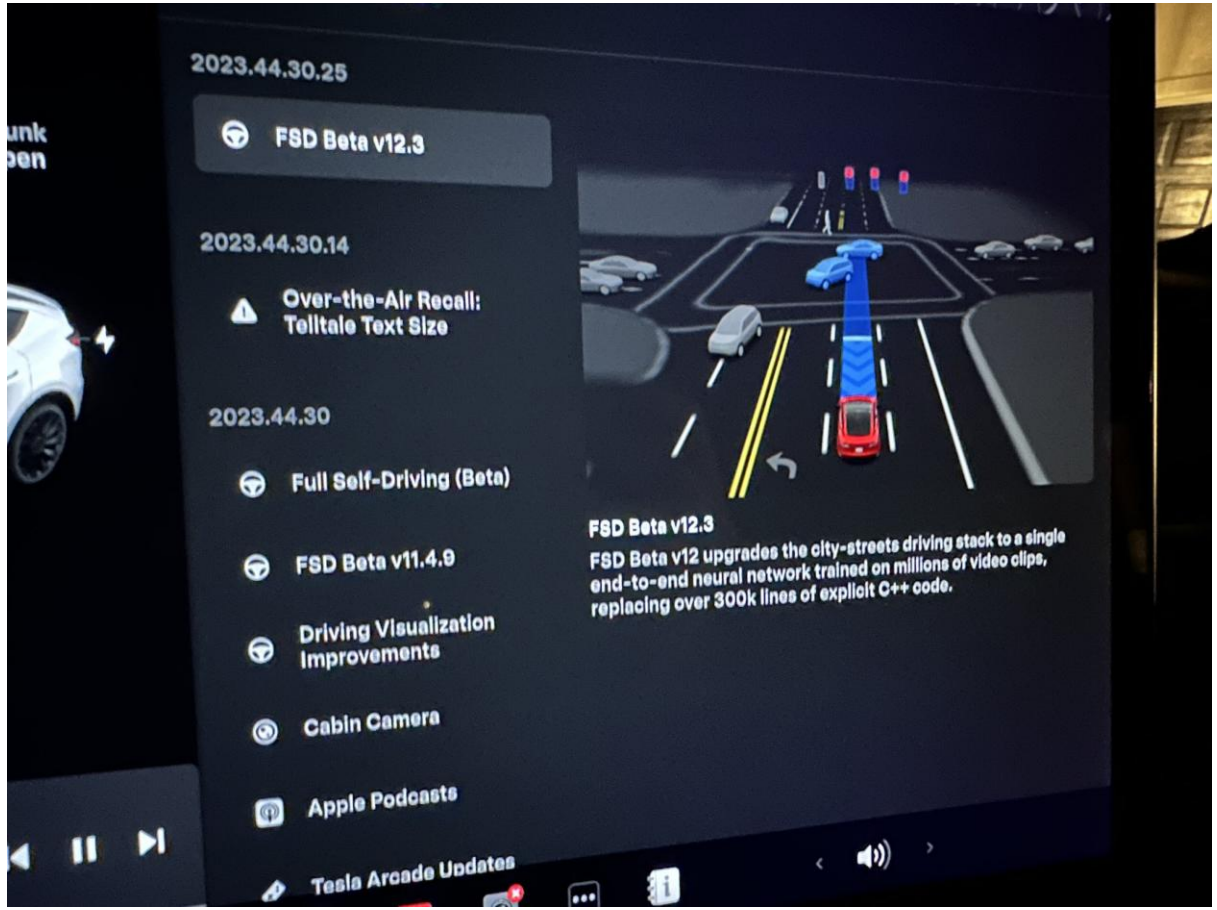


Image Source: https://x.com/uretericbud/status/1770419778340995143

Artificial Intelligence

- Through programming, humans can hand over the interaction logic designed in advance to the machine to execute repeatedly and quickly.
  - However, for tasks that require a high level of intelligence, such as face recognition, chat robots, and autonomous driving, it is difficult to design clear logic rules.
- AI is a technology that allows machines to acquire intelligent and inferential mechanisms like humans.
  - This concept first appeared at the Dartmouth Conference in 1956.
- It is undoubtedly more difficult to make intelligent machines that can reach the level of the human brain.
  - However, machines that archive similar to or even surpass human intelligence in some way have been proven to be feasible.



Image: https://www.reddit.com/r/ProgrammerHumor/comments/8owrso/if_advanced_ai_was_a_scooby_doo_villain/

Three Stages of AI Development

In the 1970s, scientists tried to implement AI through knowledge database and reasoning. They built a large and complex expert system to simulate the intelligence level of human experts.

- One of the biggest difficulties with these explicitly specified rules is that many complex, abstract concepts cannot be implemented in concrete code.

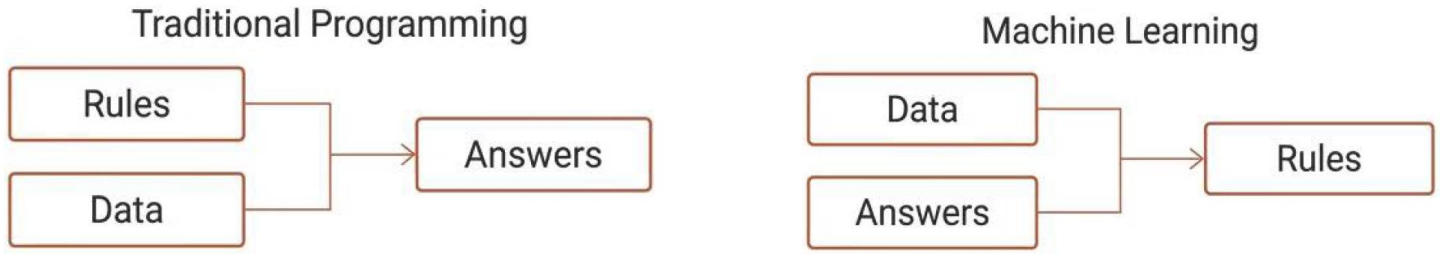
To solve such problems, a research discipline that allowed machines to automatically learn rules from data, known as machine learning, was born.

- Machine learning became a popular subject in AI in the 1980s.

Since 2012, the applications of deep neural network technology have made major breakthroughs in fields like computer vision, natural language processing (NLP), and robotics.

- Some tasks have even surpassed the level of human intelligence.
- Deep neural networks eventually have a new name – deep learning.

The Paradigm Shift



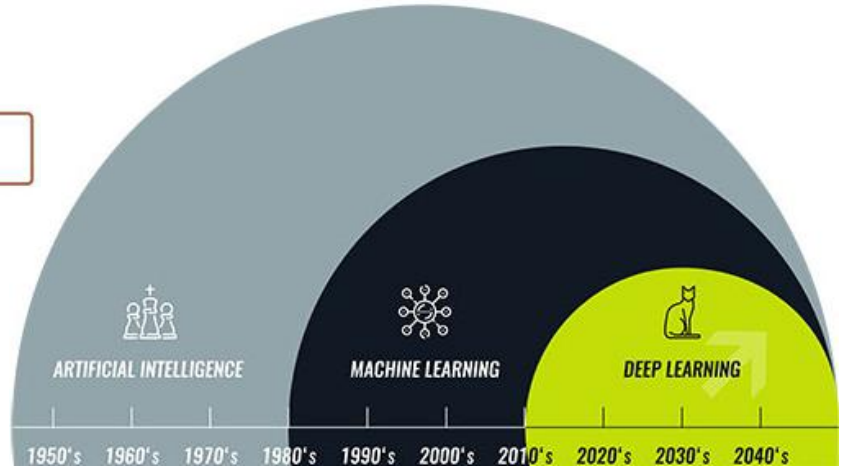


Image: https://www.twt.de/news/detail/wo-liegt-der-unterschied-zwischen-artificial-intelligence-machine-learning-und-deep-learning.html

Neural Networks and Deep Learning

Attributes	Machine Learning	Deep Learning
Data Dependency	Performs well on small/medium data	Requires large datasets (Big Data)
Hardware	Can run on CPUs	Relies on GPUs/TPUs
Feature Engineering	Manual extraction required	Automated (End-to-End)
Training Time	Fast to train	Computationally intensive

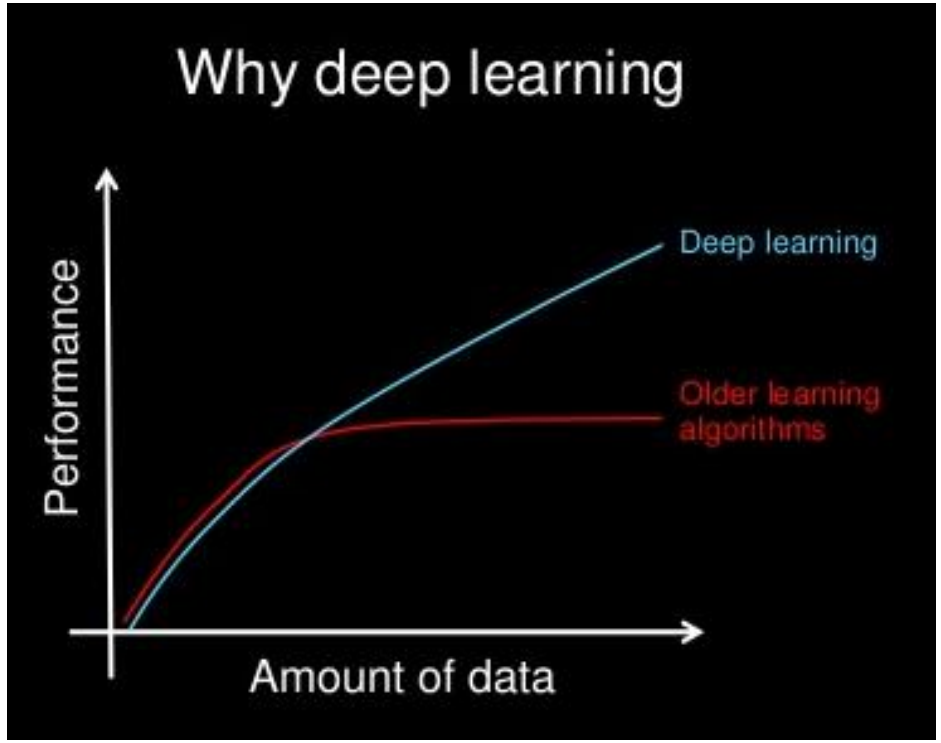


Image: https://www.width.ai/post/advantages-of-deep-learning



> I think people need to understand that deep learning is making a lot of things, behind-the-scenes, much better.
>
> Geoffrey Hinton

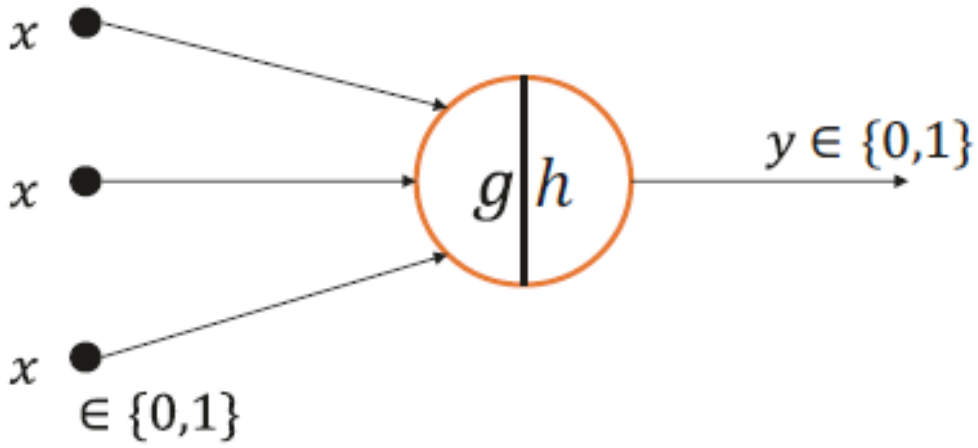
Neural Networks and Deep Learning

- In 2006, Geoffrey Hinton first proposed the concept of deep learning.
- In 2012, AlexNet, an eight-layer deep neural network, was released and achieved huge performance improvements in the image recognition competition.
- Since then, neural network models with dozens, hundreds, and even thousands of layers have been developed successively, showing strong learning ability.

Image: https://www.linkedin.com/pulse/geoffrey-hinton-alan-francis/

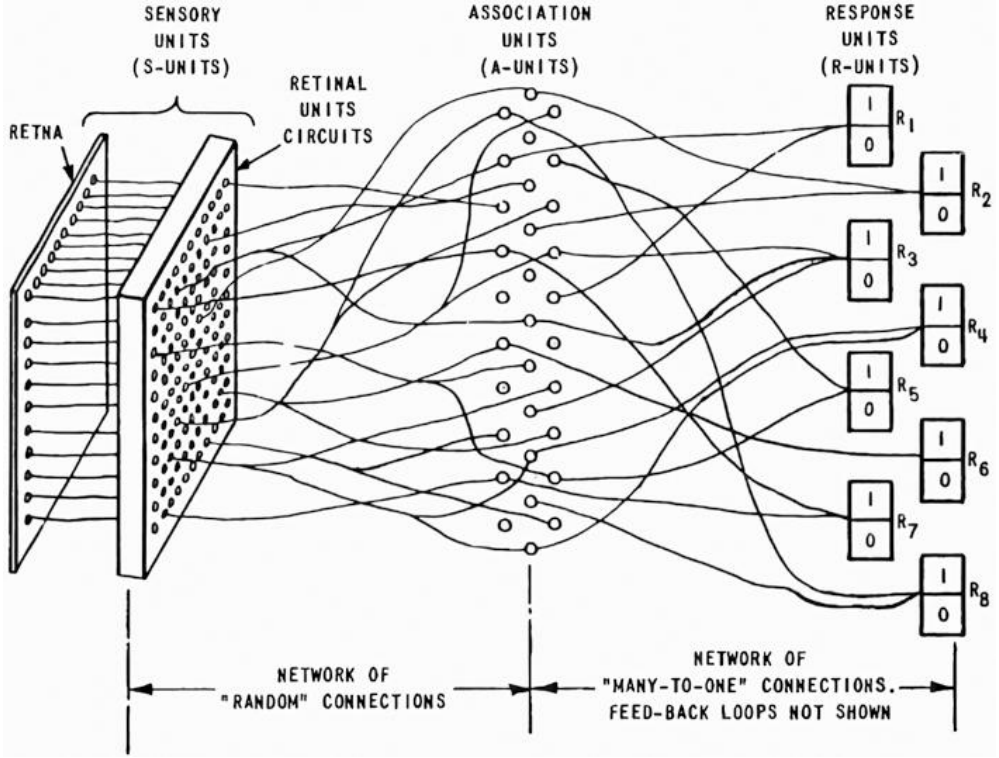
Shallow Neural Networks

- In 1943, psychologist Warren McCulloch and logician Walter Pitts proposed the earliest mathematical model of neurons based on the structure of biological neurons.
- It called MP neuron models after their last name initials.
- The MP neuron models have no learning ability and can only complete fixed logic judgments.

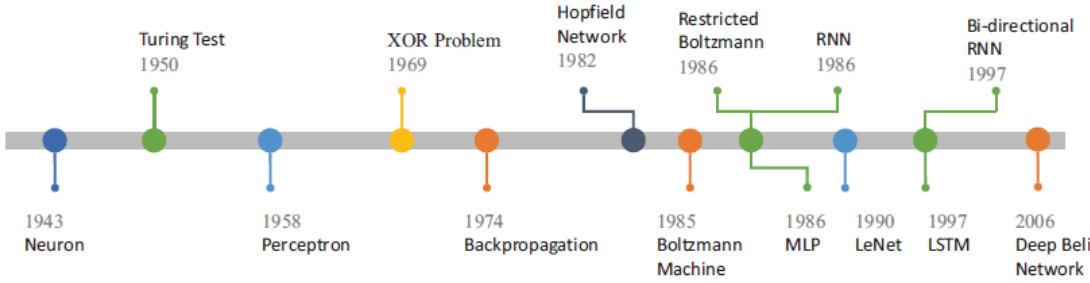


Shallow Neural Networks

- In 1958, American psychologist Frank Rosenblatt proposed the first neuron model that can automatically learn weights, called perceptron.
- Frank Rosenblatt then implemented the perceptron model based on the "Mark 1 perceptron" hardware.
- The input is an image sensor with 400 pixels, and the output has eight nodes. It can successfully identify some English letters.
- It is generally believed that 1943–1969 is the first prosperous period of artificial intelligence development.

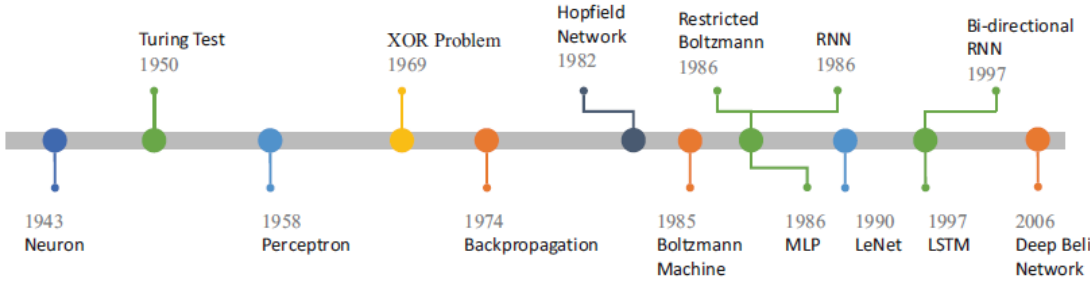


History



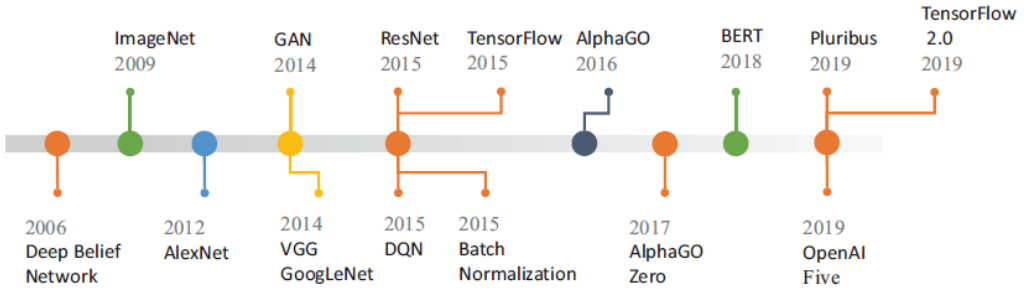
- The mathematical idea of the **backpropagation (BP)** algorithm, which is still the core foundation of modern deep learning algorithms, has been derived as early as the 1960s, but it had not been applied to neural networks at that time.
- In 1974, American scientist Paul Werbos first proposed that the **BP algorithm can be applied to neural networks** in his doctoral dissertation. Unfortunately, this result has not received enough attention.
- In 1986, David Rumelhart et al. published a paper using the **BP algorithm for feature learning** in Nature. Since then, the BP algorithm started gaining widespread attention.
- The **second wave of artificial intelligence** renaissance was started from 1982 to 1995.
  - During this period, convolutional neural networks, recurrent neural networks, and backpropagation algorithms were developed one after another.
- In 1989, Yann LeCun and others applied the BP algorithm to **handwritten digital image recognition** and achieved great success, which is known as **LeNet**.
  - LeNet was successfully commercialized in zip code recognition, bank check recognition, and many other systems.

History



- In 1997, one of the most widely used recurrent neural network variants, **Long Short-Term Memory (LSTM)**, was proposed by Jürgen Schmidhuber.
- Support vector machines (SVM) have a rigorous theoretical foundation, require a small number of training samples, and also have good generalization capabilities.
  - In contrast, neural networks lack theoretical foundation and are hard to interpret.
  - Deep networks are difficult to train, and the performance is normal.
- In 2006, Geoffrey Hinton et al. found that multilayer neural networks can be better trained through layer-by-layer pre-training and **achieved a better error rate than SVM on the MNIST** handwritten digital picture data set.
  - In that paper, Geoffrey Hinton first proposed the **concept of deep learning**.

History



- In 2011, Xavier Glorot proposed a **Rectified Linear Unit (ReLU)** activation function, which is one of the most widely used activation functions now.
- In 2012, Alex Krizhevsky proposed an eight-layer deep neural network **AlexNet**, which used the ReLU activation function and Dropout technology to prevent overfitting.
  - At the same time, it abandoned the layer-by-layer pre-training method and directly trained the network on two NVIDIA GTX580 GPUs.
- In 2014, Ian Goodfellow proposed **generative adversarial networks (GANs)**, which learned the true distribution of samples through adversarial training to generate samples with higher approximation.
- In 2016, DeepMind applied deep neural networks to the field of **reinforcement learning** and proposed the DQN algorithm.
  - It achieved a level comparable to or even higher than that of humans in 49 games in the Atari game platform.

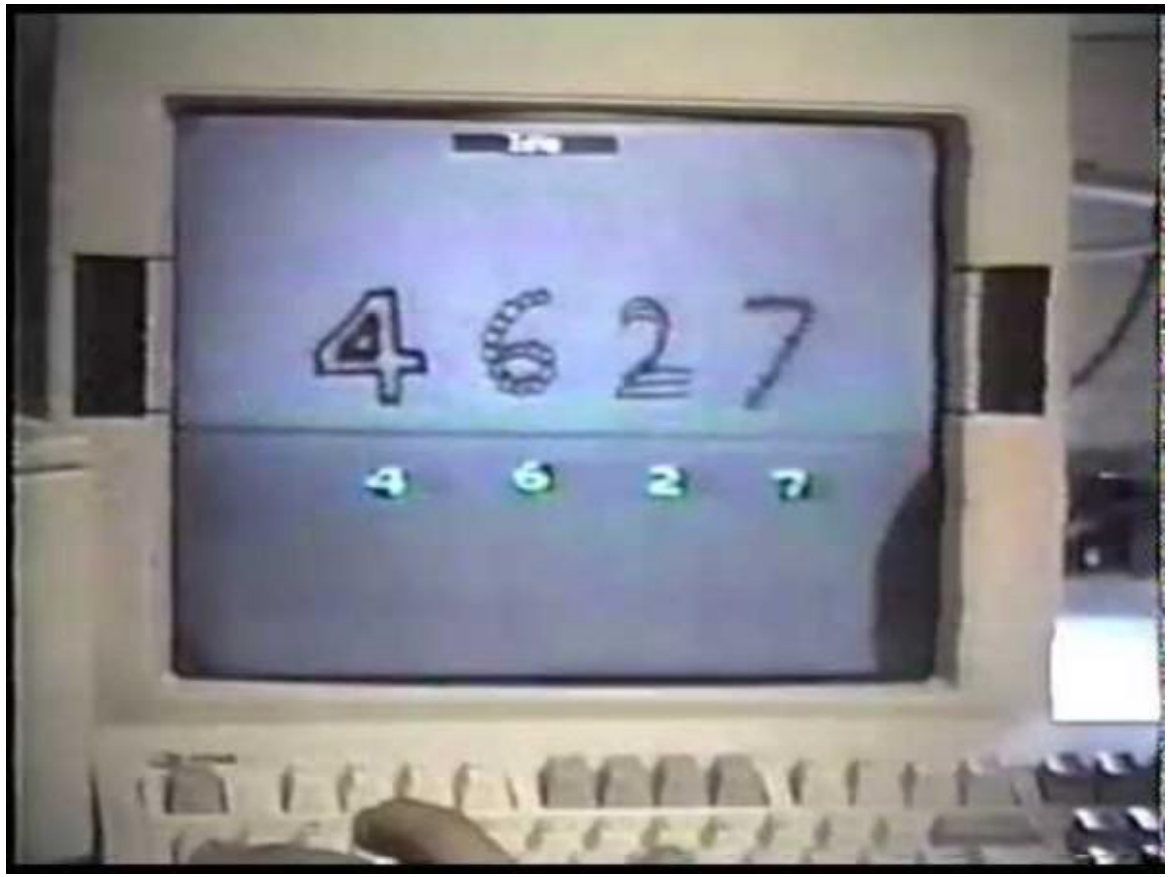
Deep Learning Characteristics

Compared with traditional machine learning algorithms and shallow neural networks, modern deep learning algorithms usually have the following characteristics:

- Data Volume
- Computing Power
- Network Scale
- General Intelligence

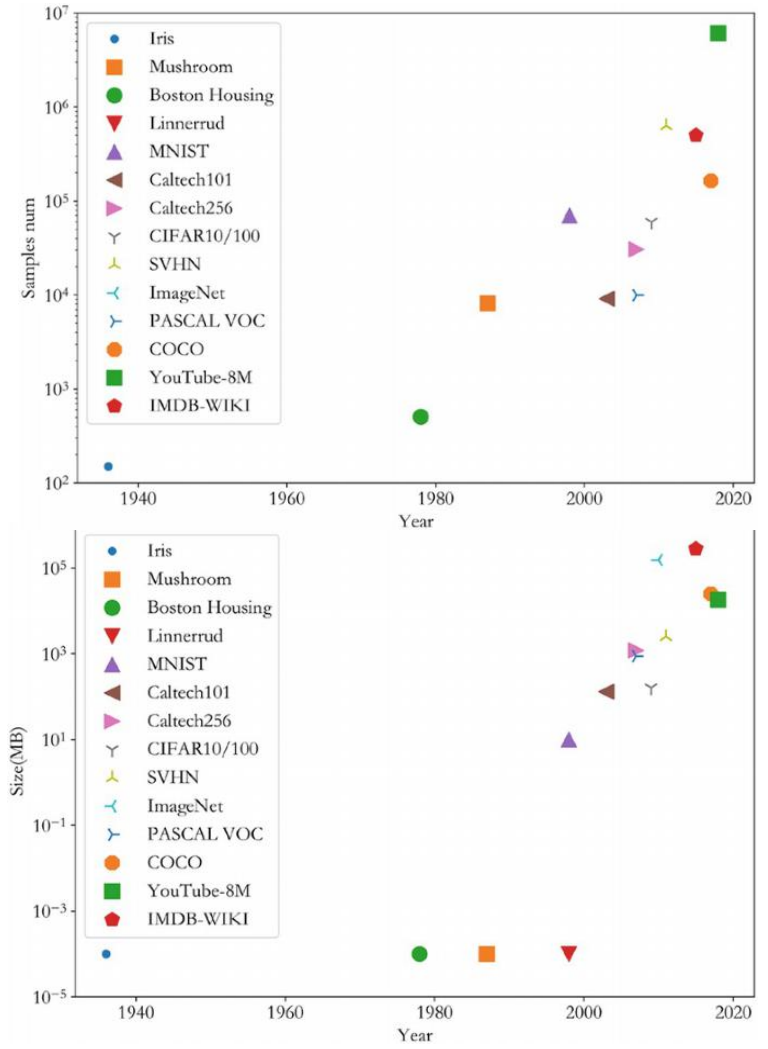
Data Volume

- Early machine learning algorithms are relatively simple and fast to train, and the size of the required dataset is relatively small.
  - The Iris flower dataset collected by the British statistician Ronald Fisher in 1936, which contains only three categories of flowers, with each category having 50 samples.
- The algorithms are more and more complex, and the demand for data volume is also increasing.
  - The MNIST handwritten digital picture dataset collected by Yann LeCun in 1998 contains a total of ten categories of numbers from 0 to 9, with up to 7,000 pictures in each category.



Data Volume

- With the rise of neural networks, especially deep learning networks, the number of network layers is generally large, and the number of model parameters can reach one million, ten million, or even one billion.
  - The ImageNet dataset released in 2010 included a total of 14,197,122 pictures, and the compressed file size of the entire dataset was 154GB.
- Collecting data, especially collecting labeled data, is often very expensive.
  - The formation of a dataset usually requires manual collection, crawling of raw data and cleaning out invalid samples, and then annotating the data samples with human intelligence.
  - Subjective bias and random errors are inevitably introduced.

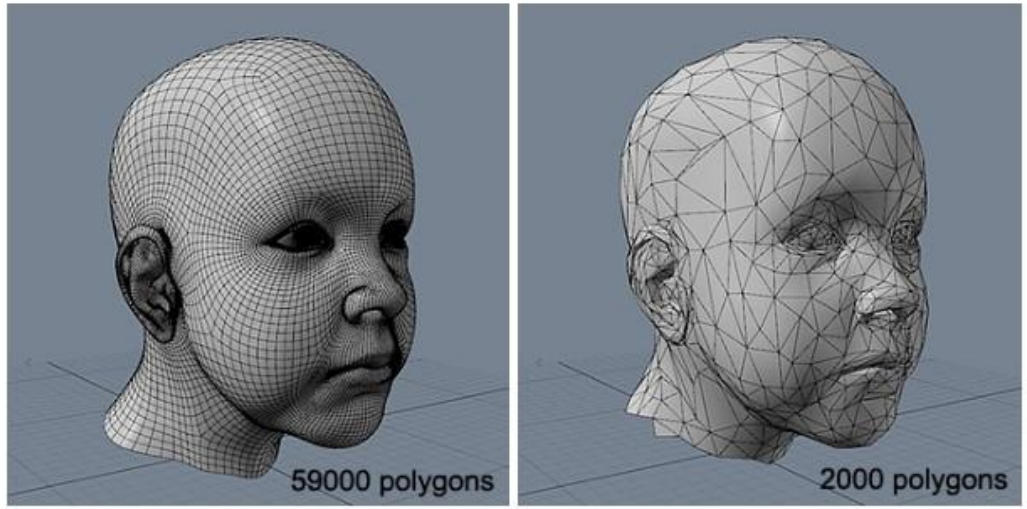


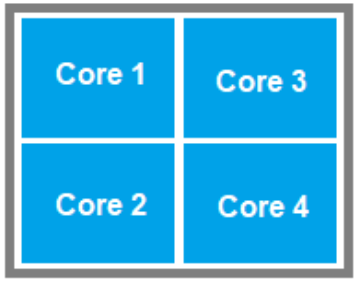
Computing Power

- The basic theory of modern deep learning was proposed in the 1980s, but the real potential of deep learning was not realized until the release of AlexNet based on training on two GTX580 GPUs in 2012.
- Traditional machine learning algorithms do not have stringent requirements on data volume and computing power like deep learning.
- Most of current neural networks use parallel acceleration chips such as NVIDIA GPU and Google TPU to train model parameters.
  - The AlphaGo Zero program needs to be trained on 64 GPUs from scratch for 40 days before surpassing all AlphaGo historical versions.
- The floating-point computing capacity of GPU grows exponentially, which is mainly driven by the increasing business of game and deep learning computing.



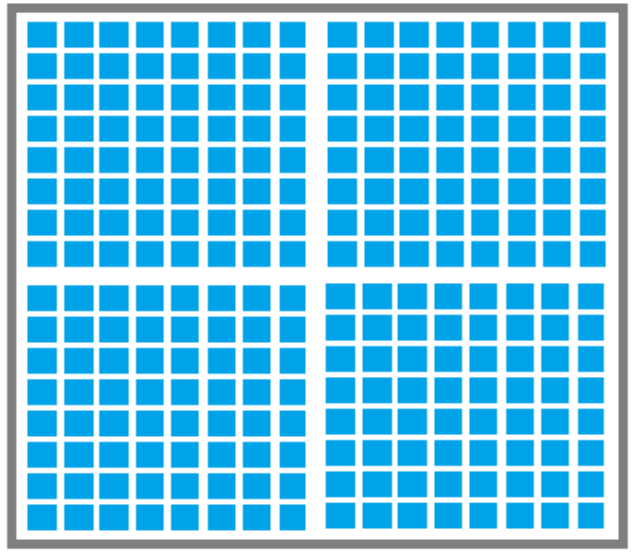
Computing Power





(Fewer strong cores)

CPU



(Thousands weaker cores)

GPU

Image Source:
https://irendering.net/how-to-optimize-gpu-usage-when-rendering-3d/
https://tecadmin.net/cpu-vs-gpu-key-differences/

Network Scale

- Early perceptron models and multilayer neural networks only have one or two to four layers, and the network parameters are also around tens of thousands.
- With the development of deep learning and the improvement of computing capabilities, models such as AlexNet (8 layers), VGG16 (16 layers), GoogleNet (22 layers), ResNet50 (50 layers), and DenseNet121 (121 layers) have been proposed successively.
- The size of inputting pictures has also gradually increased from 28×28 to 224×224 to 299×299 and even larger.
- The increase of network scale enhances the capacity of the neural networks correspondingly, so that the networks can learn more complex data modalities and the model performance can be improved accordingly.
- On the other hand, the increase of the network scale also means that we need more training data and computational power to avoid overfitting.

Network Scale



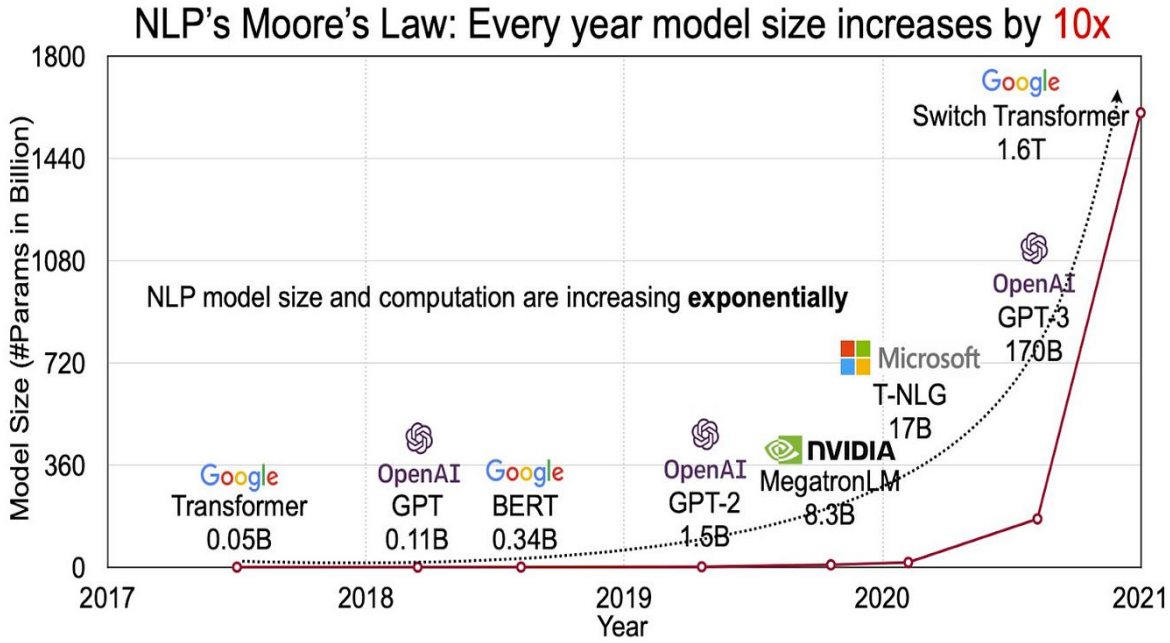


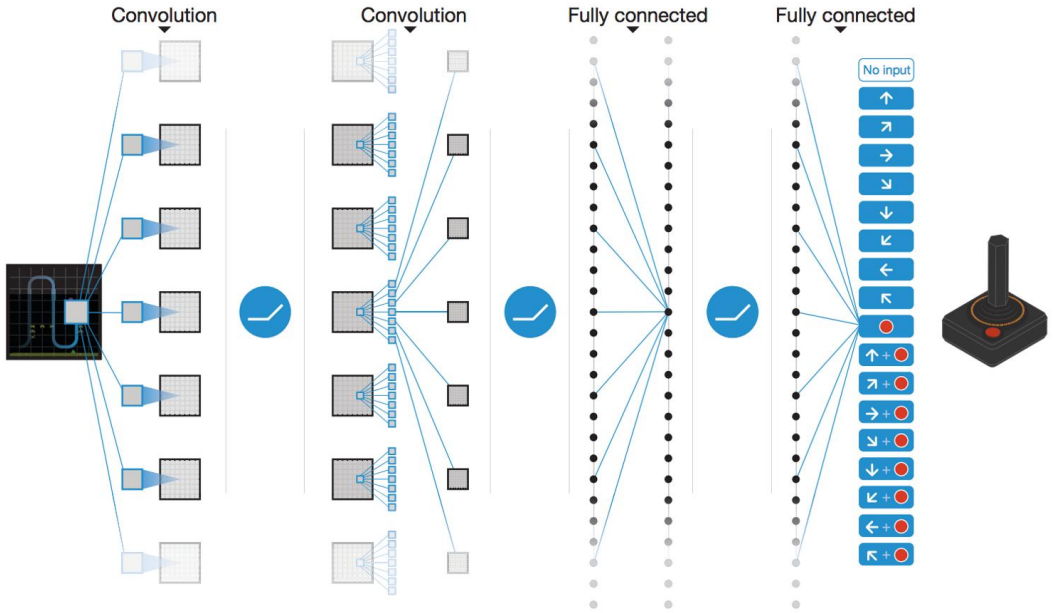
Image:
https://www.datacamp.com/blog/everything-we-know-about-gpt-5
https://medium.com/@harishdatalab/unveiling-the-power-of-large-language-models-llms-e235c4eba8a9

General Intelligence

- In the past, in order to improve the performance of an algorithm on a certain task, it is often necessary to use prior knowledge to manually design corresponding features to help the algorithm better converge to the optimal solution.
- Designing a universal intelligent mechanism that can automatically learn and self-adjust like the human brain has always been the common vision of human beings.
- Deep learning is one of the algorithms closest to general intelligence. In the computer vision field, previous methods that need to design features for specific tasks and add a priori assumptions have been abandoned by deep learning algorithms.
- At present, almost all algorithms in image recognition, object detection, and semantic segmentation are based on end-to-end deep learning models, which present good performance and strong adaptability.

General Intelligence

- On the Atari game platform, the DQN algorithm designed by DeepMind can reach human equivalent level in 49 games under the same algorithm, model structure, and hyperparameter settings.
- It is not designed for a certain game but can control 49 games on the Atari game platform.



Deep Learning Applications

Some mainstream applications of deep learning starting with:

- Computer Vision
- Natural Language Processing
- Reinforcement Learning

Computer Vision

- **Image classification** is a common classification problem.
- The input of the neural network is pictures, and the output value is the probability that the current sample belongs to each category.
- Generally, the category with the highest probability is selected as the predicted category of the sample.
- Image recognition is one of the earliest successful applications of deep learning.
- Classic neural network models include VGG series, Inception series, and ResNet series.

Image	Label
	Cat
	Cat
	Dog
	Dog

Image: https://subscription.packtpub.com/book/big-data-and-business-intelligence/9781838555078/1/ch01lvl1sec08/model-training

Computer Vision

- **Object detection** refers to the automatic detection of the approximate location of common objects in a picture by an algorithm.
- It is usually represented by a bounding box and classifies the category information of objects in the bounding box.
- Common object detection algorithms are RCNN, Fast RCNN, Faster RCNN, Mask RCNN, SSD, and YOLO series.



Image: https://towardsdatascience.com/why-is-object-detection-so-messy-6a91191579b8

Computer Vision

- **Semantic segmentation** is an algorithm to automatically segment and identify the content in a picture.
- We can understand semantic segmentation as the classification of each pixel and analyze the category information of each pixel.
- Common semantic segmentation models include FCN, U-net, SegNet, and DeepLab series.



Computer Vision

- As deep learning achieves better results on 2D picture–related tasks, 3D **video understanding** tasks with temporal dimension information (the third dimension is sequence of frames) are receiving more and more attention.
- Common video understanding tasks include video classification, behavior detection, and video subject extraction.
- Common models are C3D, TSN, DOVF, and TS_LSTM.

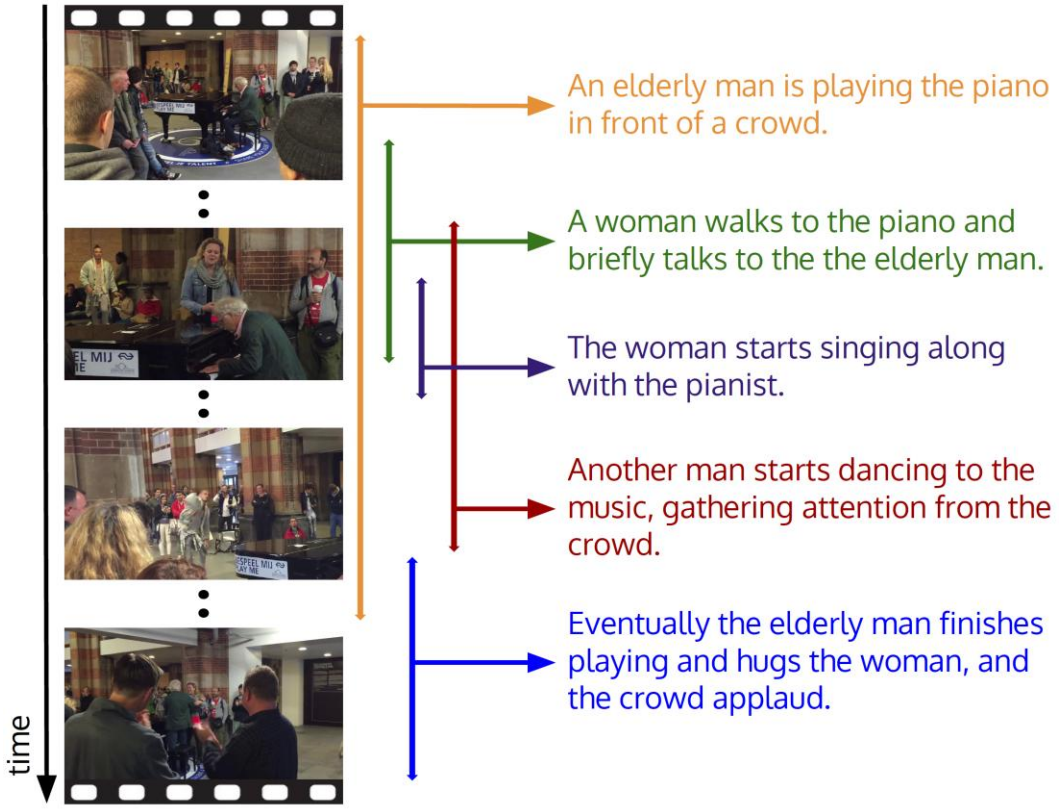


Image: https://blog.fastforwardlabs.com/2021/12/14/an-introduction-to-video-understanding-capabilities-and-applications.html

Computer Vision

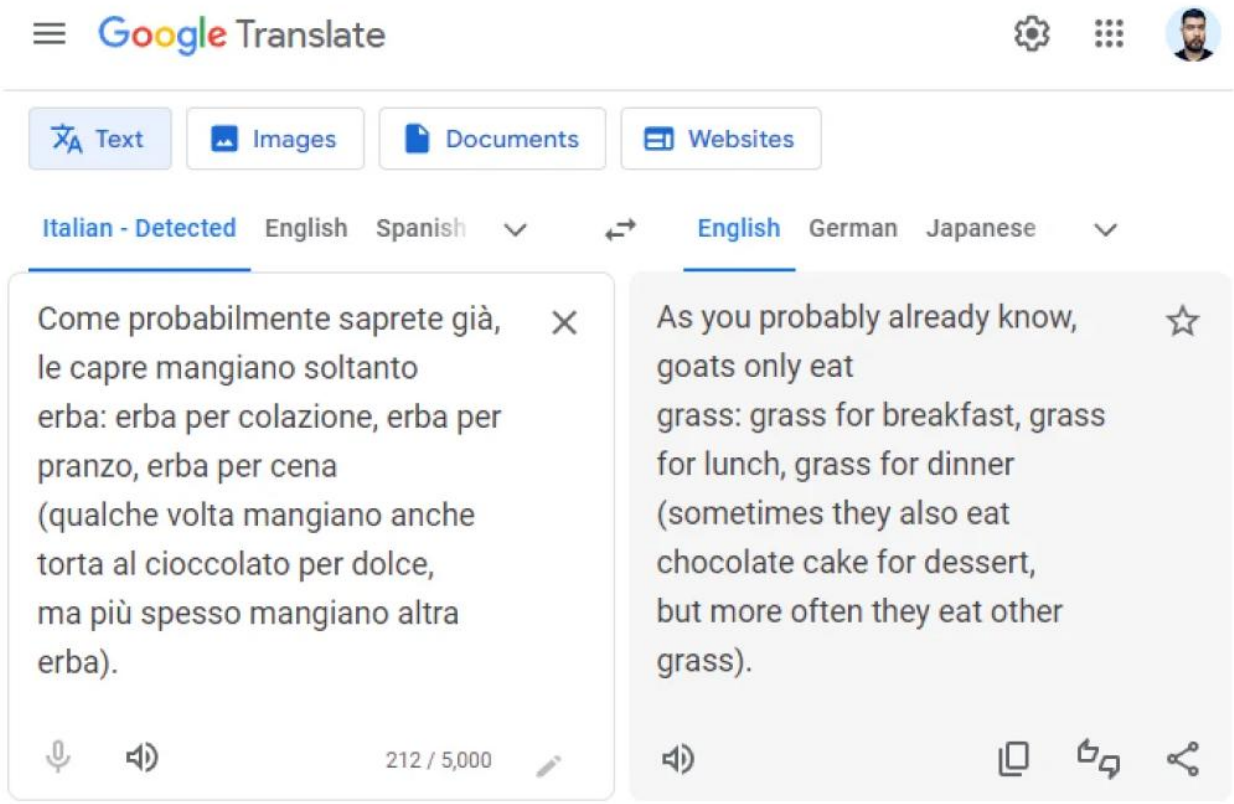
- **Image generation** learns the distribution of real pictures and samples from the learned distribution to obtain highly realistic generated pictures.
- At present, common image generation models include VAE series and GAN series. Among them, the GAN series of algorithms have made great progress in recent years.
- The picture effect produced by the latest GAN model has reached a level where it is difficult to distinguish the authenticity with the naked eye.
- In addition to the preceding applications, deep learning has also achieved significant results in other areas, such as artistic style transfer.





Natural Language Processing

- In the past, **machine translation** algorithms were usually based on statistical machine translation models.
  - It were also the technology used by Google's translation system before 2016.
- In November 2016, Google launched the Google Neural Machine Translation (GNMT) system based on the Seq2Seq model.
  - For the first time, the direct translation technology from source language to target language was realized with 50–90% improvement on multiple tasks.



Natural Language Processing

- A **chatbot** that is not only able to answer questions impressively
  - But also admit its mistakes and is able to reject requests that are not appropriate.
  - Moreover, ChatGPT remembers the prompt used in the conversation.
- Examples:
  - Write poetries or even songs (even in the style of a specific singer or author).
  - **Write** headlines, jokes (or explanations), cover letters, or even short essays.
  - Provide **suggestions** for projects or ideas for books.
  - Generate code (or debug it).
  - Do recommendations (e.g., books, movies, series), and **summarization** (e.g., Youtube videos, books)

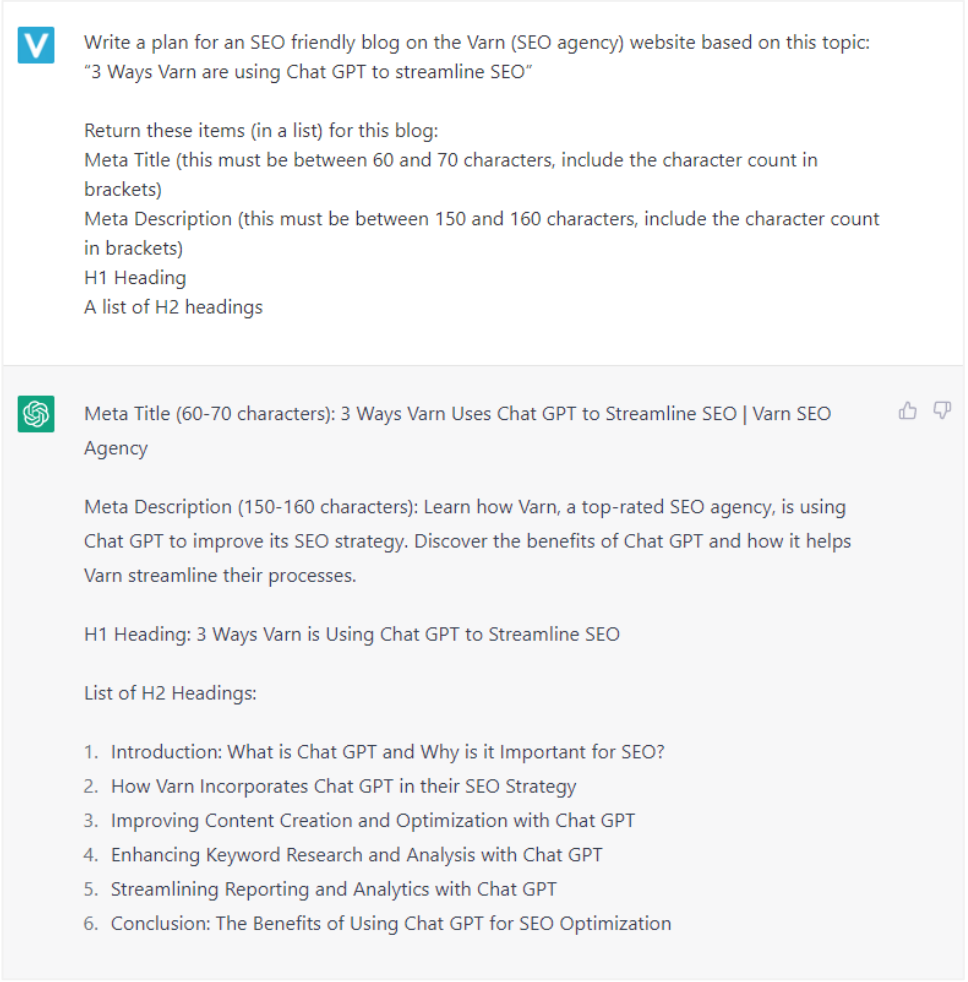


Image: https://varn.co.uk/03/23/3-ways-varn-are-using-chat-gpt-to-streamline-seo/

Reinforcement Learning

- Compared with the real environment, **virtual game** platforms can
  - train and test reinforcement learning algorithms
  - avoid interference from irrelevant factors
  - while minimizing the cost of experiments
- Commonly used virtual game platforms include OpenAI Gym, OpenAI Universe, OpenAI Roboschool, DeepMind OpenSpiel, and MuJoCo.
- Commonly used reinforcement learning algorithms include DQN, A3C, A2C, and PPO.
- In Dota 2 and StarCraft games, AI developed by OpenAI and DeepMind have defeated professional teams under restriction rules.

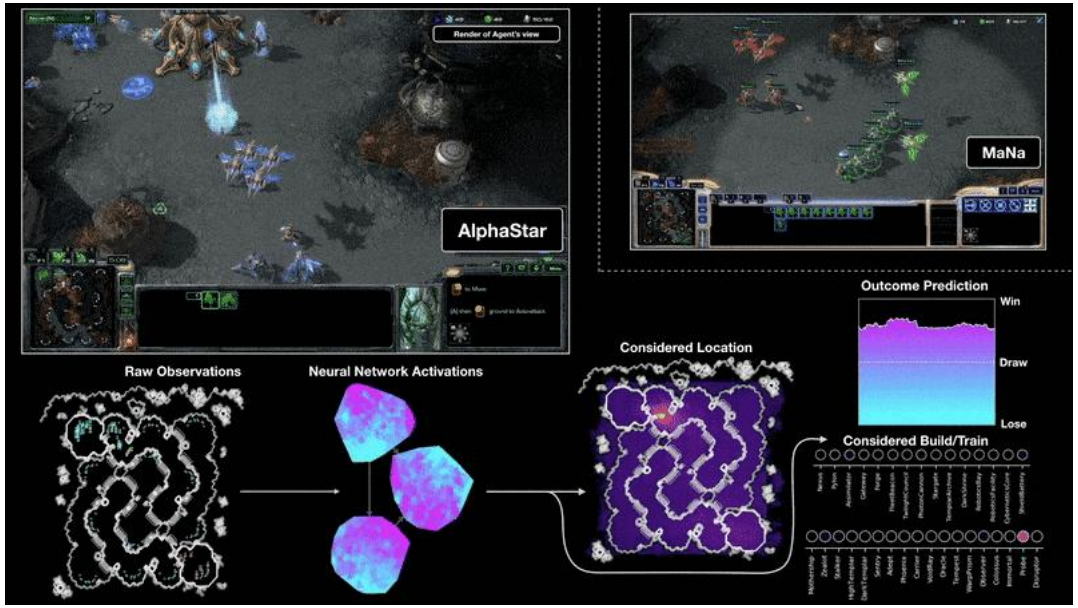


Image: https://www.deepmind.com/blog/alphastar-mastering-the-real-time-strategy-game-starcraft-ii

Reinforcement Learning

- In the real environment, the control of robots has also made some progress.
- For example, UC Berkeley Lab has made a lot of progress in the areas of imitation learning, meta learning, and few-shot learning in the field of **robotics**.
- Boston Dynamics has made gratifying achievements in robot applications. The robots it manufactures perform well on tasks such as complex terrain walking and multi-agent collaboration.



Image: https://www.kedglobal.com/future-mobility/newsView/ked202012080011

Reinforcement Learning



Video: https://youtu.be/7TKP3YRbVmM?si=y1rl6ZRtFK-vy5Nu

Reinforcement Learning

- **Autonomous driving** is considered as an application direction of reinforcement learning in the short term.
- Many companies have invested a lot of resources in autonomous driving, such as Tesla, Uber, and Google.
- Car manufactures equip some autonomous features in most of their cars nowadays.



Video:https://www.tesla.com/de_DE/videos/autopilot-self-driving-hardware-neighborhood-short

Reinforcement Learning



Video: https://www.youtube.com/watch?v=O8TSA-X9UlU

Deep Learning Framework

- **Scikit-learn** is a complete computing library for machine learning algorithms.
  - It has built-in support for common traditional machine learning algorithms, and it has rich documentation and examples.
- However, scikit-learn is not specifically designed for neural networks.
  - It does not support GPU acceleration, and the implementation of neural network–related layers is also lacking.

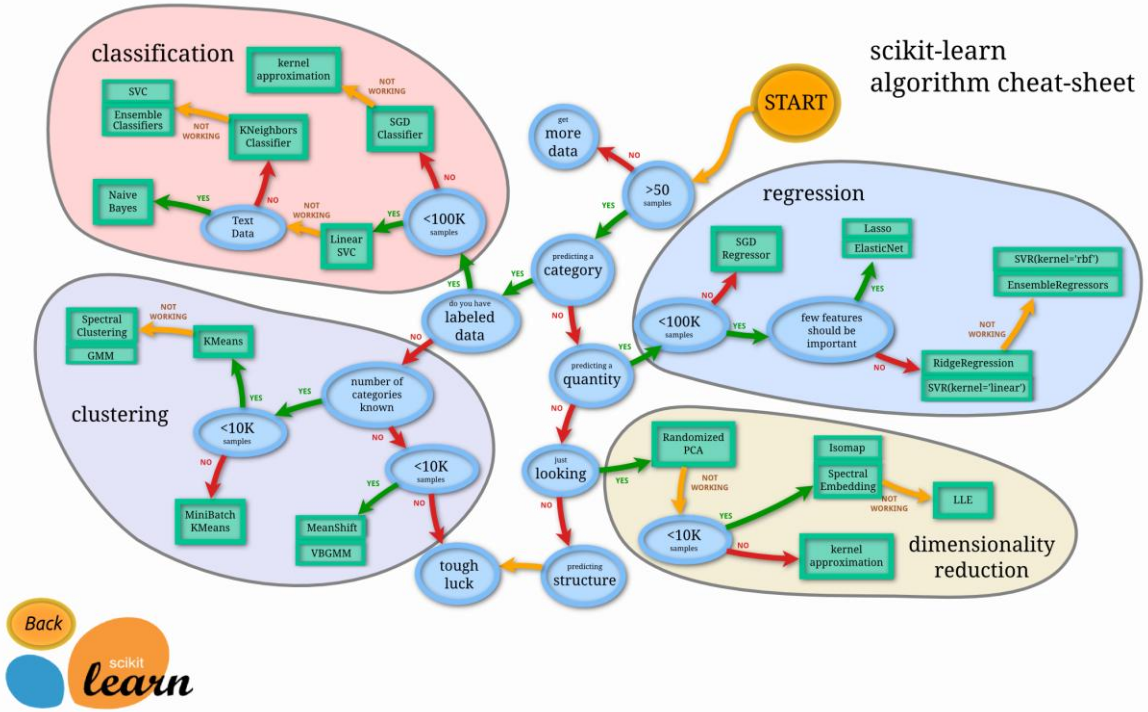


Image : https://scikit-learn.org/stable/tutorial/machine_learning_map/index.html

Deep Learning Framework

- **PyTorch** is a deep learning framework launched by Facebook based on the original Torch framework using Python as the main development language.
  - PyTorch borrowed the design style of Chainer and adopted imperative programming, which made it very convenient to build and debug the network.
- Although PyTorch was only released in 2017, due to its sophisticated and compact interface design, PyTorch has received wide acclaim in the academic world.
- After the 1.0 version, the original PyTorch and Caffe2 were merged to make up for PyTorch's deficiencies in industrial deployment.
- Overall, PyTorch is an excellent deep learning framework



Image: https://www.rapidminer.jp/course/1191/

Deep Learning Framework

- **Keras** is a high-level framework implemented based on the underlying operations provided by frameworks such as Theano and TensorFlow.
- It provides a large number of high-level interfaces for rapid training and testing.
- For common applications, developing with Keras is very efficient.
  - But because there is no low-level implementation, the underlying framework needs to be abstracted, so the operation efficiency is not high, and the flexibility is average.



Image: https://keras.io/

Deep Learning Framework

- **TensorFlow** is a deep learning framework released by Google in 2015.
  - The initial version only supported symbolic programming.
- Thanks to its earlier release and Google's influence in the field of deep learning, TensorFlow quickly became the most popular deep learning framework.
  - However, due to frequent changes in the interface design, redundant functional design, and difficulty in symbolic programming development and debugging, TensorFlow 1.x was once criticized by the industry.
- In 2019, Google launched the official version of TensorFlow 2, which runs in dynamic graph priority mode and can avoid many defects of the TensorFlow 1.x version.
  - TensorFlow 2 has been widely recognized by the industry.

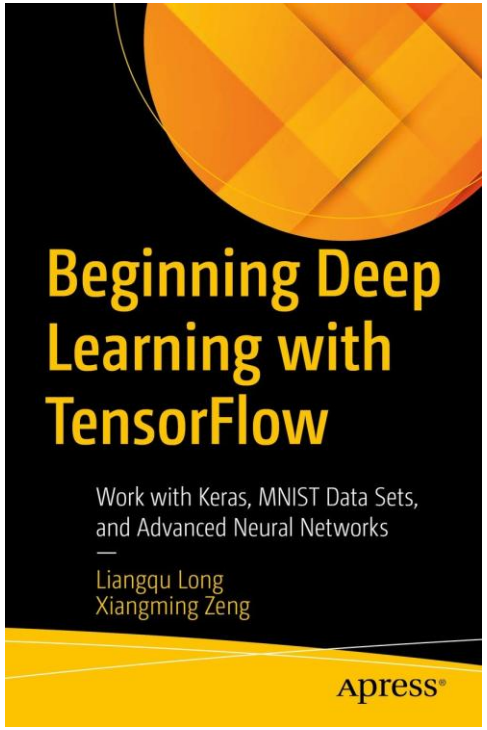


Image: https://datasolut.com/einfuehrung-in-tensorflow/

Deep Learning Framework

- At present, **TensorFlow** and **PyTorch** are the two most widely used deep learning frameworks in industry.
- TensorFlow has a complete solution and user base in the industry.
- Thanks to its streamlined and flexible interface design, PyTorch can quickly build and debug networks, which has received rave reviews in academia.
- After TensorFlow 2 was released, it makes it easier for users to learn TensorFlow and seamlessly deploy models to production.

References


Beginning Deep Learning with TensorFlow
Work with Keras, MNIST Data Sets, and Advanced Neural Networks
Liangqu Long
Xiangming Zeng
Apress

Vielen Dank für deine Aufmerksamkeit!

Kontakt:
Prof. Dr.-Ing. Binh Vu
Fakultät für Information, Medien und Design
SRH Hochschule Heidelberg
Ludwig-Guttmann-Straße 6
69123 Heidelberg
T +49 6221 6799 - 208
binh.vu@srh.de
www.hochschule-heidelberg.de